Lab Report: Fine-tuning a Transformer for Patent Phrase Matching

Notebook link (final submission): kaggle.com/code/krrishsingh06/patent-phrase-final

Task 1: Fine-tuned baseline

The starter notebook (DeBERTa-v3-small cross-encoder, 3 epochs, LR 2e-5) gave a **validation Pearson of 0.8201** and trained in **5.6 minutes** (337.9 s) on a Kaggle GPU. When I ran the same notebook again with the same seed, it gave **0.8140** (5.5 min). So the score changes by about 0.006 between runs even when nothing is changed.

Epoch	Val Pearson (run 1)	Val Pearson (run 2)
1	0.7961	0.7870
2	0.8173	0.8098
3	0.8201	0.8140

Why the split holds out whole anchors: each anchor appears with many different targets. If we split random rows, the same anchor would be in both training and validation, so the model would already have seen it. The validation score would then be too high and would not show how the model does on new anchors.

Note: I replaced `warmup_ratio=0.1` with the same 10% warmup given as `warmup_steps`, because the installed transformers version no longer accepts `warmup_ratio`. Nothing else in the starter was changed.

Task 2: Baseline without fine-tuning

Fine-tuning added about **0.23** in Pearson over a pretrained model used with no training.

Model	Trained on this data?	Val Pearson
Bi-encoder: all-MiniLM-L6-v2, cosine similarity of anchor and target	No	0.5814
Cross-encoder: DeBERTa-v3-small (starter, same split)	Yes	0.8140

Bi-encoder vs cross-encoder: a bi-encoder turns the anchor and the target into two separate vectors and then compares them with cosine similarity. The two phrases never look at each other inside the model. A cross-encoder reads both phrases and the context as one input, so attention can compare them word by word.

What fine-tuning added: the pretrained bi-encoder only knows general meaning and ignores the patent context. The fine-tuned cross-encoder learns what "similar" means in this dataset and compares the two phrases directly, which is why it scores much higher.

Task 3: Controlled experiments

Neither experiment clearly beat the baseline. Each run changed one setting from the starter and used the same split and seed.

Setting changed	Val Pearson	Train time (min)	Interpretation
None (baseline)	0.8140 (0.8201 in another run)	5.5	Reference. Two identical runs differ by 0.006, so this is the noise level.
No context (target only)	0.8137	5.1	Almost no change; the one-word section name adds very little.
Learning rate 2e-5 → 5e-5	0.8205	5.5	+0.0065, but within run-to-run noise, so not a clear gain.

Conclusion: removing the context made no real difference, so the CPC section name gives the model little extra information here. The higher learning rate looked slightly better, but the baseline itself moved by 0.006 between two runs with the same seed, so this gap may be noise. More seeds would be needed to be sure. With LR 5e-5 the training loss went much lower (0.030 vs 0.041) while validation loss rose in epoch 3, which hints at some overfitting.

Task 4: Error analysis

All numbers below use the baseline model's `val_predictions.csv` (val Pearson 0.8140).

1. Pearson per CPC section

Section	Field	Rows	Pearson
B	Performing operations; transporting	1926	0.8912

Section	Field	Rows	Pearson
F	Mechanical engineering	968	0.8410
E	Fixed constructions	373	0.8061
H	Electricity	933	0.7957
A	Human necessities	943	0.7797
D	Textiles; paper	199	0.7765
C	Chemistry; metallurgy	1177	0.7561
G	Physics	1040	0.7023

The hardest sections are **G (Physics)** and **C (Chemistry)**. Physics phrases are often short and general (like "imaging axis"), so small word changes can change the meaning a lot. Chemistry uses special names and short forms the model may not know well. D has only 199 rows, so its score is less reliable.

2. Mean prediction for each true score

True score	0.00	0.25	0.50	0.75	1.00
Mean prediction	0.118	0.310	0.499	0.676	1.029
Rows	1522	2465	2475	832	265

Yes, the model pulls some scores toward the middle. Pairs with score 0 get 0.118 on average (too high) and pairs with 0.75 get 0.676 (too low). Most labels are 0.25 or 0.5, so the loss pushes predictions toward these common values. Score 1.0 is not pulled in, probably because those pairs are almost the same phrase and easy to spot.

3. Ten largest errors

Anchor	Target	Context	True	Pred
punch face	face punch	B05	0.00	0.878
boom hydraulic cylinder	boom a hydraulic cylinder	E02	0.00	0.838
imaging axis	imaging of axis	G02	0.00	0.765
end pins	wood pin	G10	0.75	-0.011
activating position	active region	G06	0.00	0.685

Anchor	Target	Context	True	Pred
imaging axis	image of axis	G02	0.00	0.675
breech mechanism	breech loading naval gun	B29	0.75	0.080
return structure	return to structure	D01	0.00	0.668
travele in direction	travele in response	F42	0.00	0.656
boom hydraulic cylinder	boom a hydraulic	E02	0.00	0.619

Failure types:

1. **Same words, different meaning (8 of 10).** The target uses almost the same words in a different order or with a small word added ("punch face" vs "face punch", "return structure" vs "return to structure"). The model sees the shared words and predicts high, but the label is 0.
2. **Related meaning, few shared words (2 of 10).** "end pins" vs "wood pin" and "breech mechanism" vs "breech loading naval gun" are related in their field, but share little text. The model needs field knowledge it does not have, so it predicts near 0.

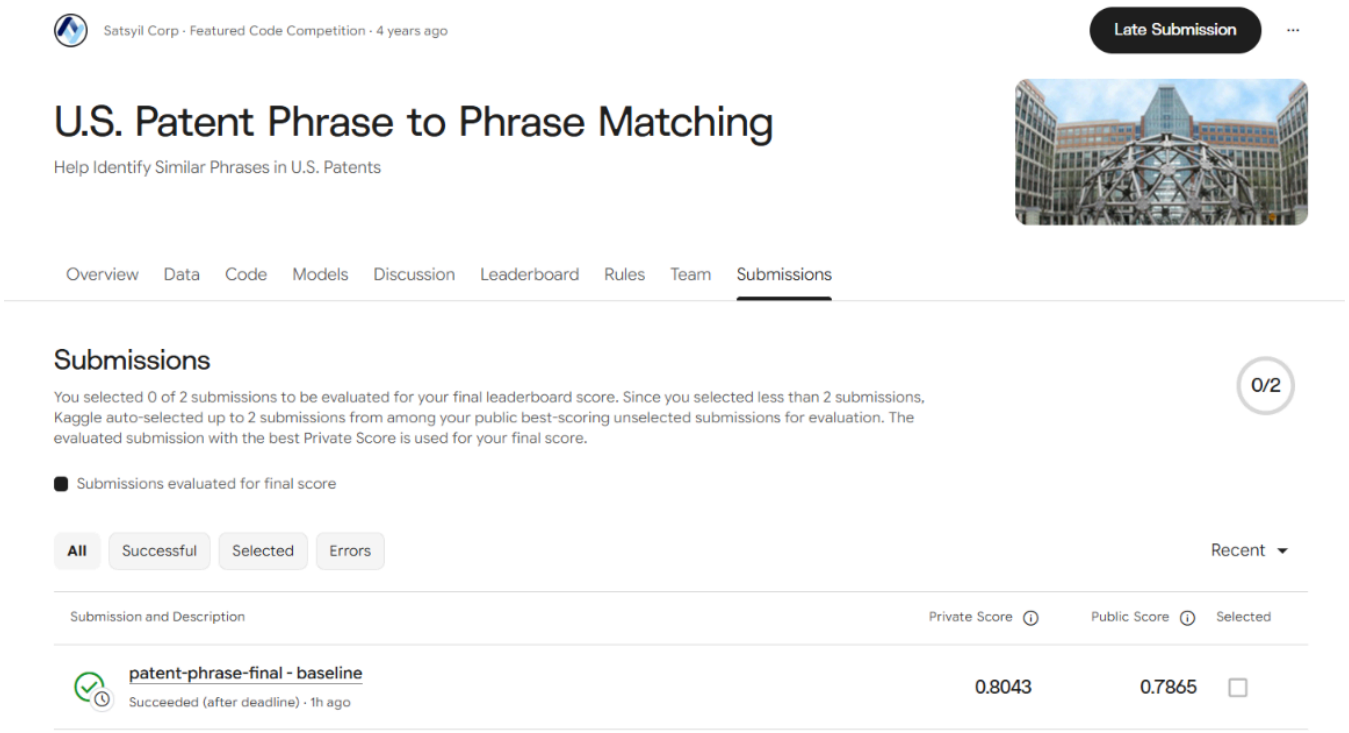
Debatable labels: "imaging axis" vs "imaging of axis" and "boom hydraulic cylinder" vs "boom a hydraulic cylinder" are rated 0, but they look very close in meaning. The model's high score here seems reasonable, so some of these "errors" may be label noise.

Task 5: Late submission and leaderboard scores

My final model is the **starter baseline**, because no experiment beat it on validation by more than the run-to-run noise. I made **one** late submission.

Submission	Validation	Public	Private
patent-phrase-final (baseline)	0.8201	0.7865	0.8043

Figure 1 (below): Kaggle Submissions tab showing this submission (Private 0.8043, Public 0.7865).



Comparison: the scores are close but not the same. Private is 0.016 below validation, and public is 0.034 below. Possible reasons:

- The hidden test set has a different mix of anchors and CPC sections than my validation split.
- The public part is smaller, so its score is noisier.
- The validation score itself moves by about 0.006 between identical runs.

References

1. Aslanyan, G., & Wetherbee, I. (2022). *Patents Phrase to Phrase Semantic Matching Dataset.* arXiv:2208.01171
2. Google Research (2022). *Announcing the Patent Phrase Similarity Dataset.* Blog post
3. Kaggle. *U.S. Patent Phrase to Phrase Matching* competition. Competition page
4. He, P., Gao, J., & Chen, W. (2021). *DeBERTaV3.* arXiv:2111.09543
5. Reimers, N., & Gurevych, I. (2019). *Sentence-BERT.* EMNLP 2019. arXiv:1908.10084
6. Hugging Face model card: sentence-transformers/all-MiniLM-L6-v2 (used in Task 2).
7. Hugging Face documentation: Trainer, Sentence Transformers.
8. Course starter notebook `patent_phrase_starter.ipynb` and `download_weights.ipynb`.